

DIJKSTRA-ALGORITHM

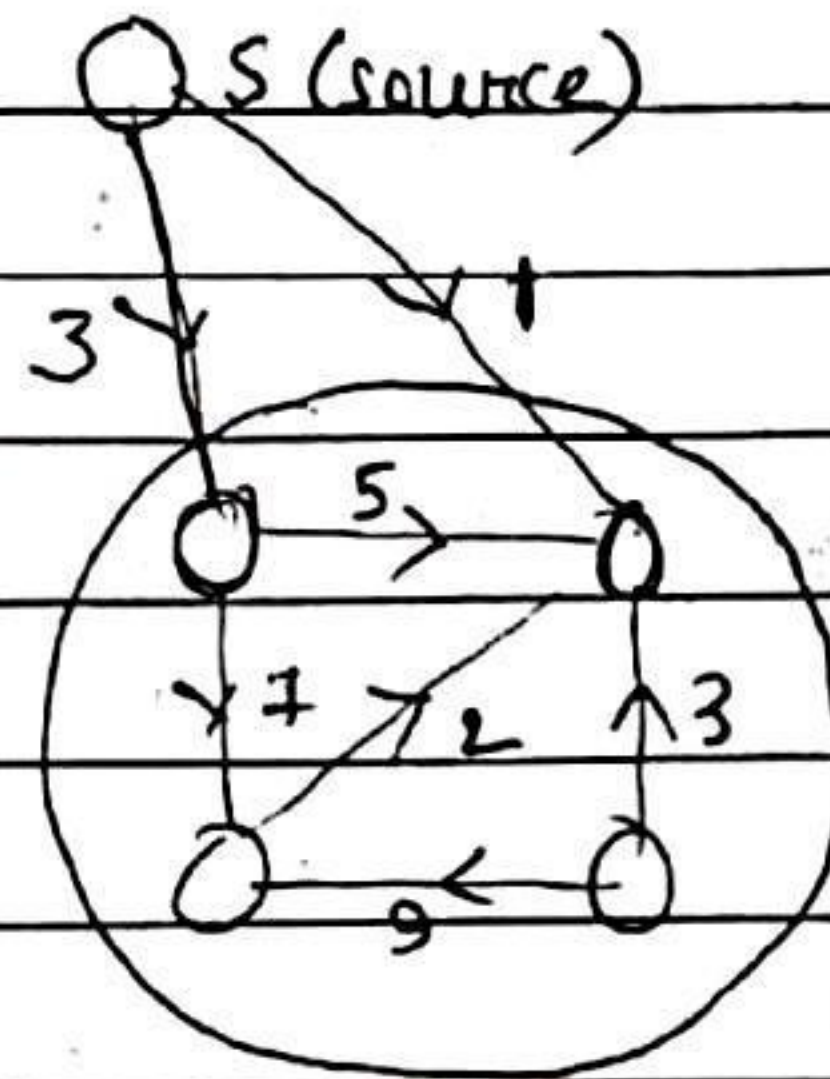
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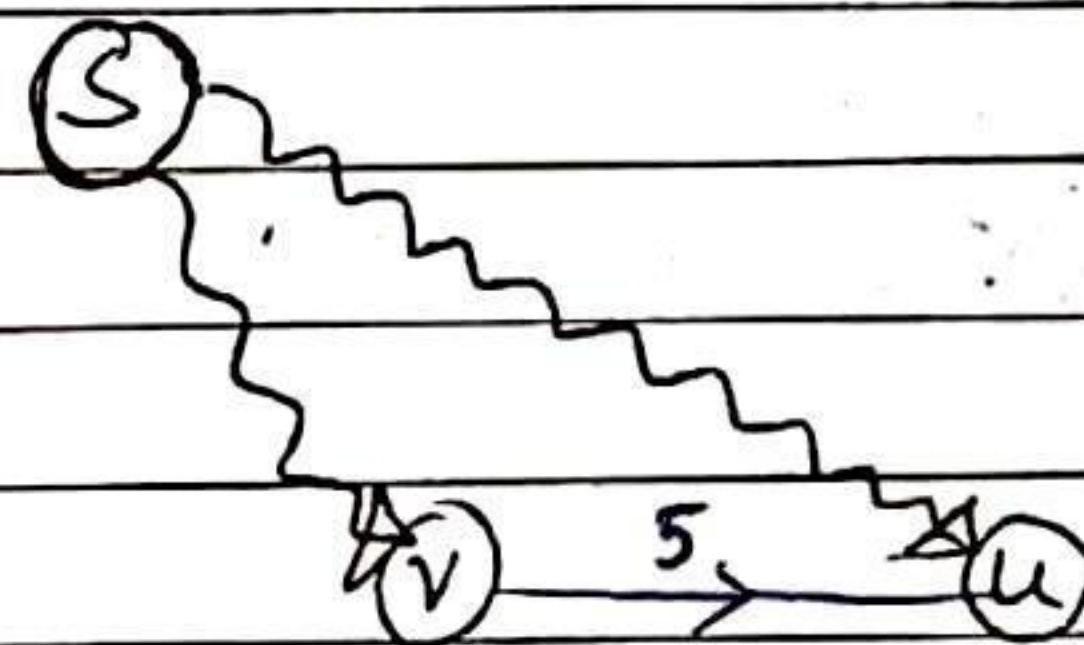
finding

• Shortest path problem (using Dijkstra Algorithm) =



find shortest path from source to others node.

Relaxing an edge (used in Dijkstra algo) =
(operation) (v, u)



$d(v) = 10$ $d(u) = 20$ 15

$$\begin{aligned} & \text{if } d(u) > d(v) + c(v, u) \\ & d(u) = d(v) + c(v, u) \\ & = 10 + 5 \\ & = 15 \end{aligned}$$

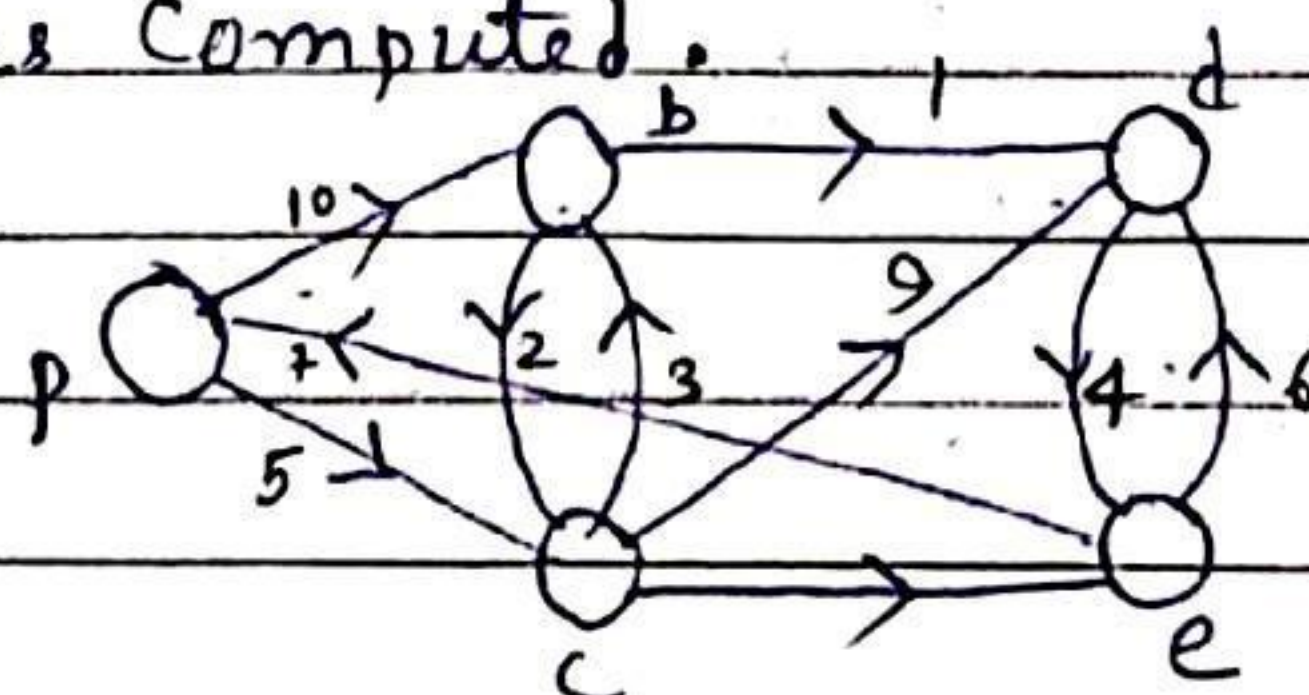
→ Dijkstra or single source shortest path algorithm is
- we are going to start from source. and
- we want to find out shortest path to all other nodes
In the graph it is call single source shortest path.

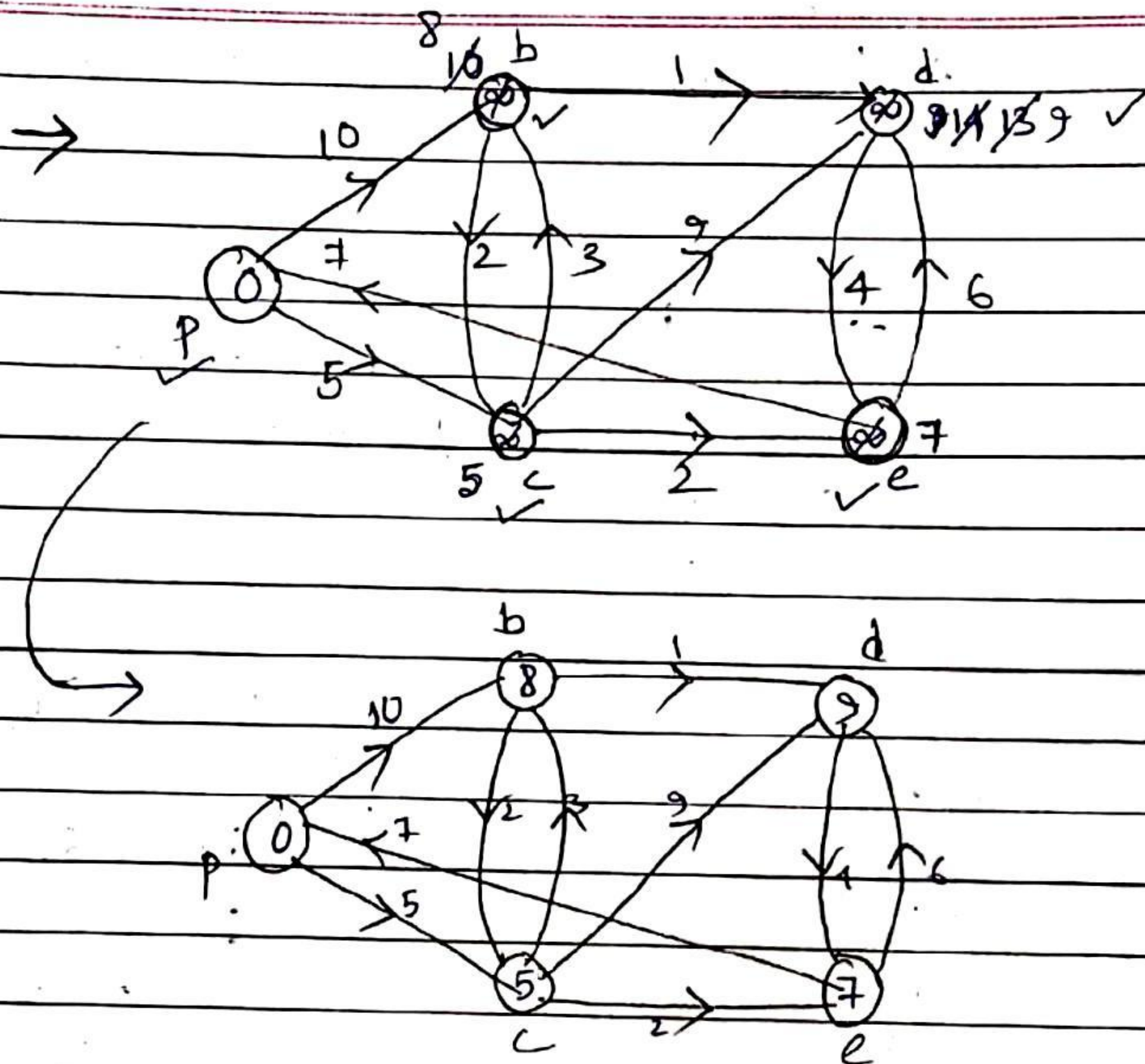
→ Dijkstra algo not applicable for negative edges.

Example-1

If you start from 'p' as source for Dijkstra algorithm what is the order in which the shortest path to other vertices computed.

given





~~p to b~~

~~p to (p, d) shortest path =~~

(p to b) shortest path = 8

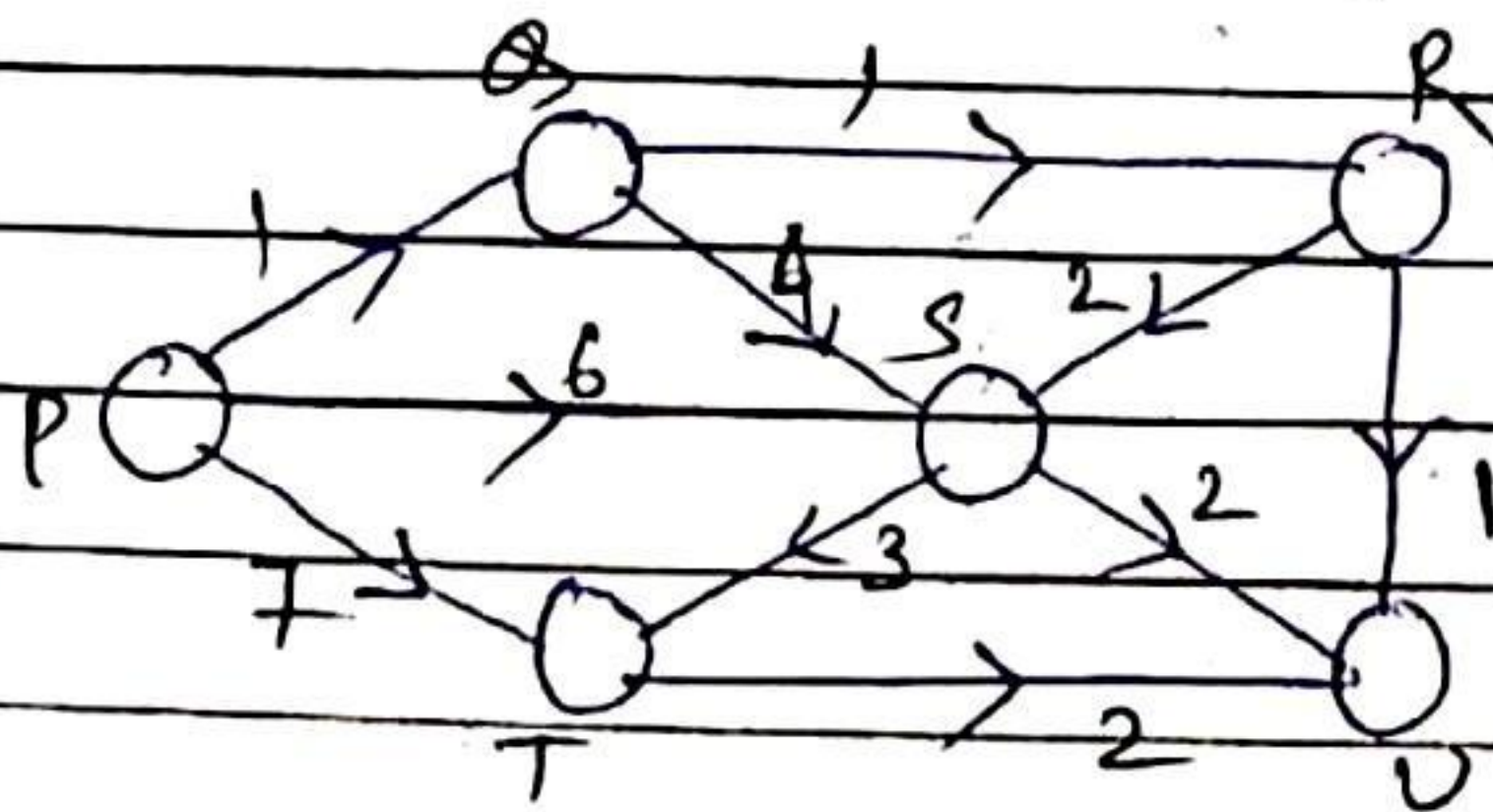
(p to c) " " = 5

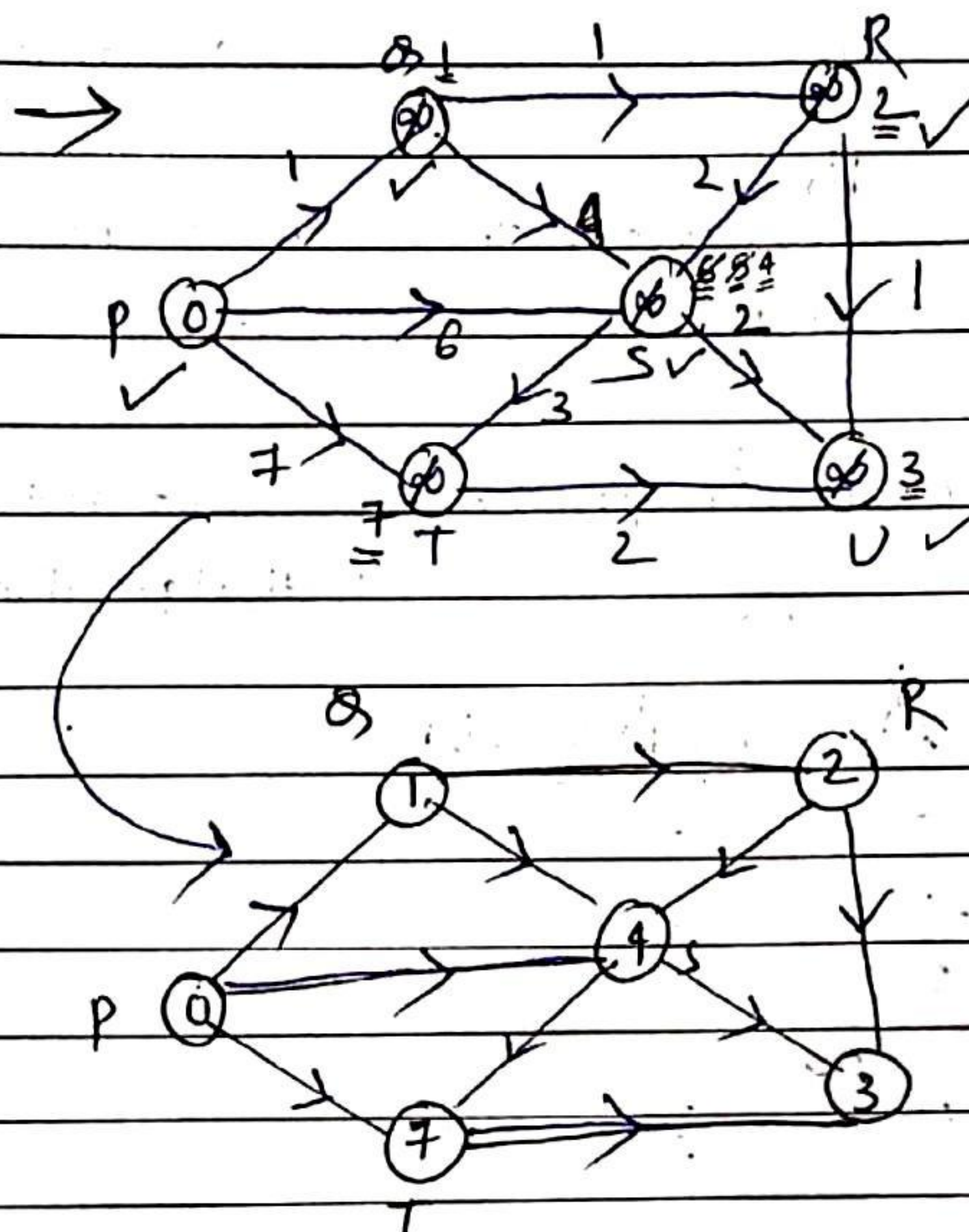
(p to d) " " = 9

(p to e) " " = 7

Example-2

Start from p as source for dijkstra algorithm then what is the order in which the shortest path to the other vertices computed - given, ~~graph~~ directed weighted graph -





→ In a undirected graph
relax all vertices.

→ In directed graph
relax all outgoing
vertices.

CRUST

• Dijkstra algorithm =

$\text{Dijkstra}(G, w, s)$
 Graph weight matrix
 Source
 Set of vertices

1. Initialize - single-source $(G, s) \rightarrow O(V)$

2. $S = \emptyset$

3. $Q = G.V$ - (Build heap) $\rightarrow O(V)$

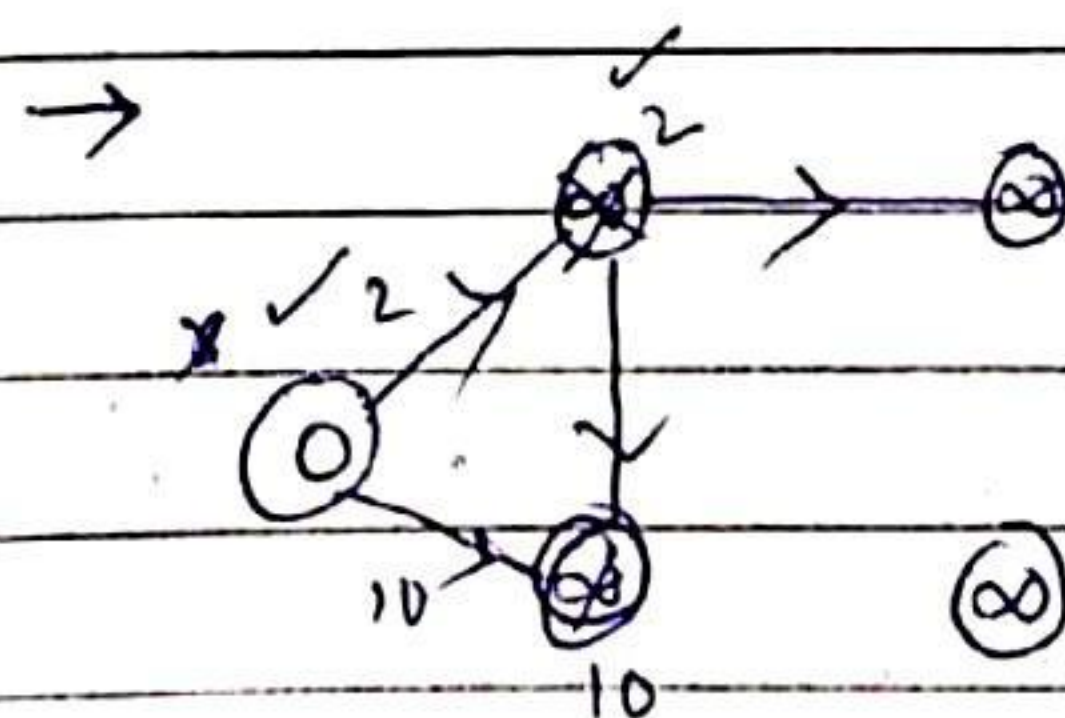
4. While $Q \neq \emptyset$ $\rightarrow V \text{ times}$

$u = \text{Extract-min}(Q) \rightarrow O(V) O(\log V)$

$S = S \cup \{u\}$

for each vertex $v \in G \text{ Adj}[u]$

Relax $(u, v, w) \rightarrow \text{Decrease key } O(\log V)$



extract-min $= O(\log V) * V$

Decrease Key $= O(\log V) * E$

$O(V + V + V \log V + E \log V)$

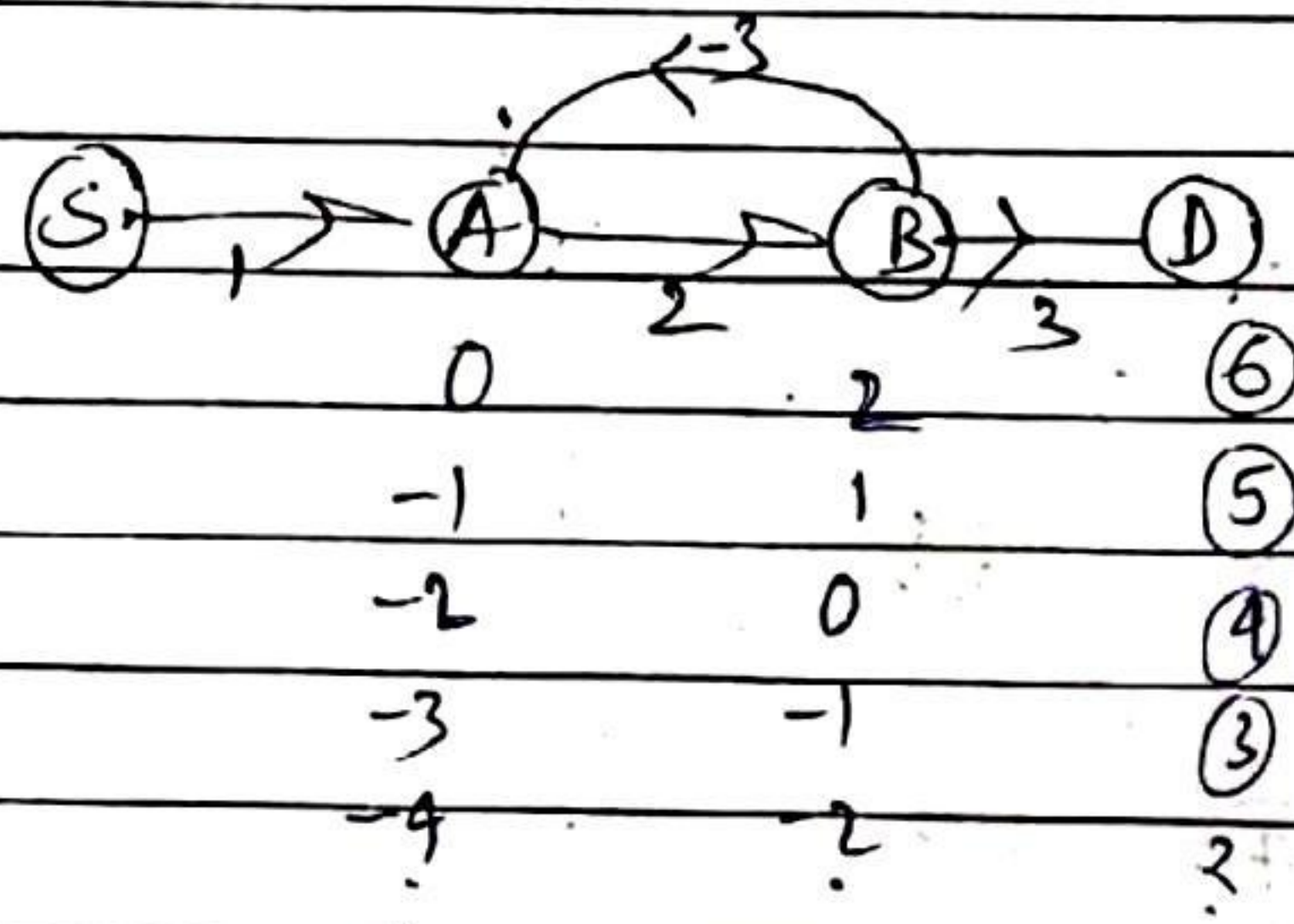
$\rightarrow O(E \log V)$

$\rightarrow$ Time complexity (Dijkstra algo using min heap)

• Negative weight edges and Dijkstra algorithm -

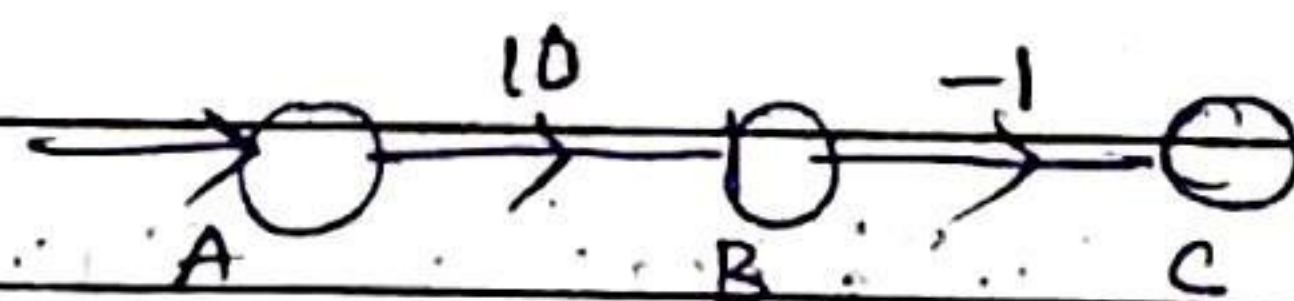
→ When graph is ^{dis}connected then we can't apply shortest path algorithm. (any algorithm)

→ ^{weight}-ve edge cycle in graph (we can't apply shortest path algorithm)

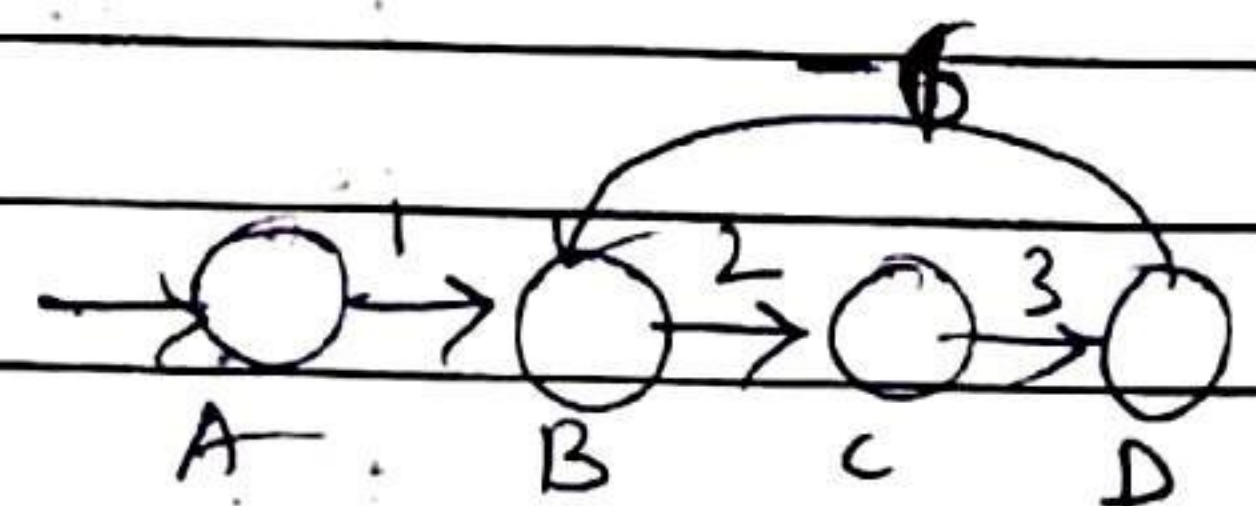
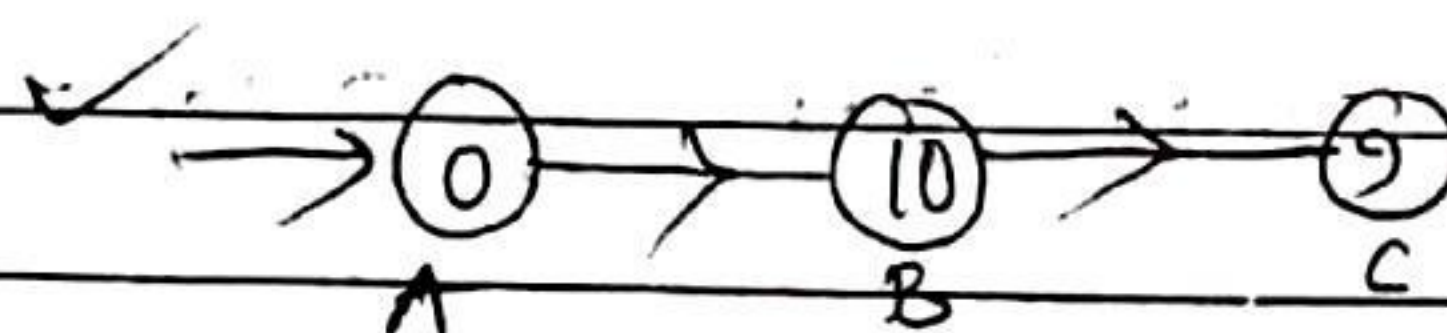


→ Dijkstra algo have not capacity to differentiate between negative ^{weight} edge and negative edge cycle.

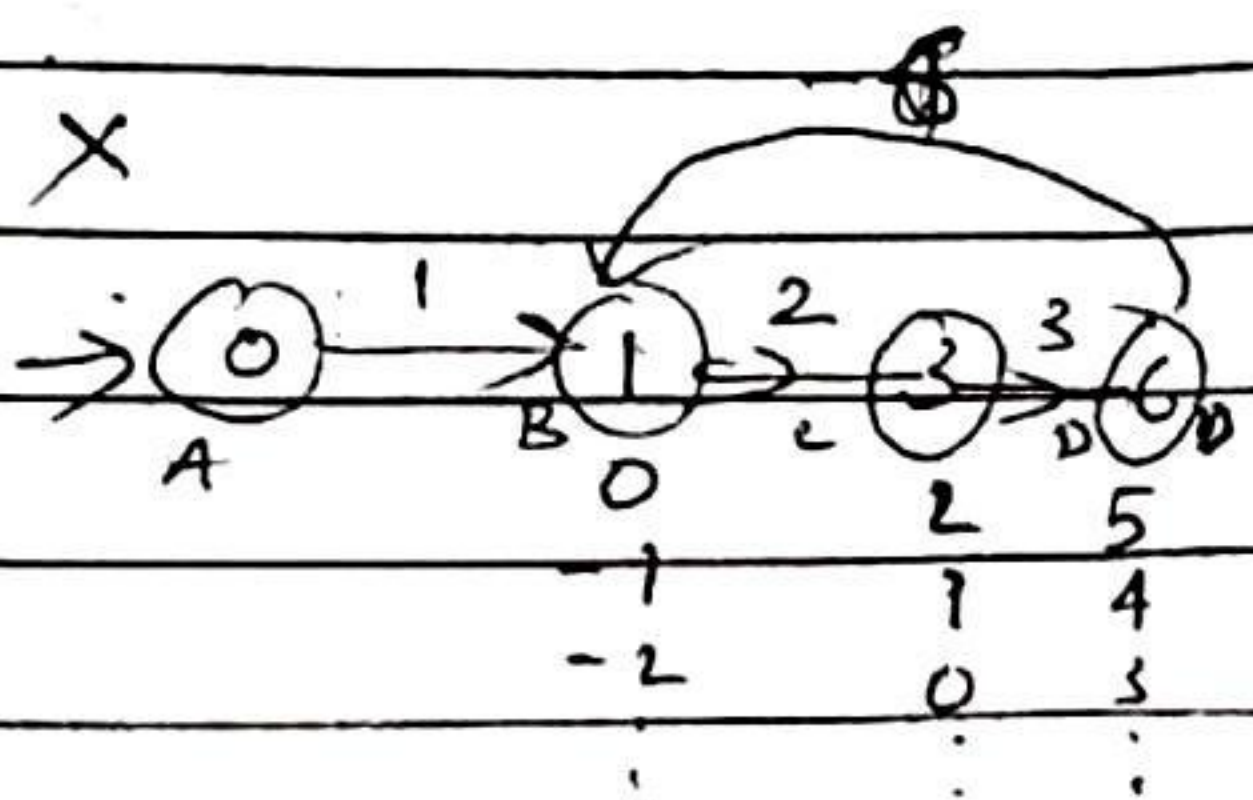
ex 1



A	B	C
0	∞	∞
	10	9



A	B	C	D
0	∞	∞	∞
	1	∞	∞
		3	∞
			6



→ Dijkstra algorithm works fine for some example where there are no -ve edge cycle.

